H1 LoRA: Final Inference Correction

Companion to TDSC manuscript v0.9 | Post-audit statistical reconciliation | 9 November 2026

Scope and immutability

All calculations use archived matched-prompt output vectors from three six-adapter libraries. The originally frozen P1 and P2 protocols and saved generations were not edited. Historical preregistered outcomes and post-audit robustness analyses are reported separately.

Reconciled state counts

Library / decision	POSITIVE	NEGATIVE	INCONCLUSIVE
P1 original (individual 95%)	12	3	42
P1 adjusted 243	10	3	44
P1 full-margin 243	8	3	46
L2 original (individual 95%)	16	24	17
L2 adjusted 243	16	23	18
L2 full-margin 243	16	23	18
L3 original (individual 95%)	16	27	14
L3 adjusted 243	14	27	16
L3 full-margin 243	9	27	21

Counts exclude the seven reference cells per library. The original counts are historical; the adjusted counts are new post-audit sensitivity results.

Corrected multiplicity and decision rule

There are 57 multi-adapter compositions in each six-adapter library. Comparing each to the base and to each active singleton produces 243 paired contrasts. The two-sided Bonferroni/Newcombe critical value at alpha=0.05 is z=3.711836 (the original reported z=3.195 is incorrect for this family). Positive status requires estimated excess at least 0.15 and positive adjusted lower bounds against all references. Negative status requires adequate observability and functionality plus at least one adjusted upper bound below 0.15. The stronger full-margin sensitivity requires every adjusted lower bound to exceed 0.15. Newcombe intervals remain approximate.

Additional robustness and hypotheses

A 729-contrast family spanning all three libraries yields 9, 16 and 14 positives (P1, L2, L3); the planted pairs remain positive. A conservative Hoeffding sensitivity, conditional on independent prompts, yields 5, 8 and 8 positives for nonzero excess, again retaining all planted pairs. Under its full-0.15-margin condition, only L2's planted pair qualifies. Frozen detection succeeds in L2 and L3; the preregistered utility requirement and combined primary hypothesis fail in L3, and both development gates failed. Historical L2/L3 closure is 15/15; under the adjusted 243-contrast procedure it is 15/15 and 13/15, respectively.

Semantic scoring and screening

The prior CPU structural audit examined 19,200 saved utility outputs and categorized 393 semantic-only acceptances, identifying 54 incomplete or malformed JSON-like outputs. Withholding structural credit altered two singleton functionality labels but did not change any audited security verdict or planted-pair utility. Retrospective replay on three cached lattices separately reports observing a positive witness and testing the exact planted pair; it is not prospective performance evidence.

Interpretation for manuscript v1.0

Preserve the original tables and figures as frozen historical output, add an explicit post-audit statistical sensitivity table and adjust the abstract, methods, results and threats-to-validity passages. Do not describe the 243 or 729 comparison family as originally preregistered, or the approximate Bonferroni/Newcombe limits as exact familywise guarantees. The study supports the existence of composition-dependent evaluation failures, not their frequency in public adapter libraries.

Artifacts: MANUSCRIPT_RECONCILIATION.md; final_inference.py; inference_summary.json; decisions_reconciled.csv; contrasts_243.csv.